

Automata Section B Quiz 1 Solution

Q1:

Step 1:

The smallest even palindrome is the empty string Λ (length 0, which is even).

Step 2:

If $x \in \text{EVEN_PALINDROME}$, then for any symbol $a \in \Sigma$ $\wedge$ $a \in \Sigma$,

$axa \in \text{EVEN_PALINDROME}$

$bxb \in \text{EVEN_PALINDROME}$

(because putting the same letter at the beginning and end keeps the word a palindrome and increases its length by 2).

Step 3:

No other strings are in EVEN_PALINDROME except those built by applying Step 2 repeatedly to Step 1.

Example with $\Sigma = \{a, b\}$

Λ (empty string)

aa, bb (from Λ)

abba, baab (from aa and bb)

aabbba, abb bba, ... (longer even palindromes)

So $\text{EVEN_PALINDROME} = \{\Lambda, aa, bb, abba, baab, aabbba, abb bba, \dots\}$

Q2:

R.E = $(a+b)^*a(a+b)(a+b)$

Q3:

R.E = $(1+10)^*$

Q4:

R.E = $(00)^* 1(11)^*$